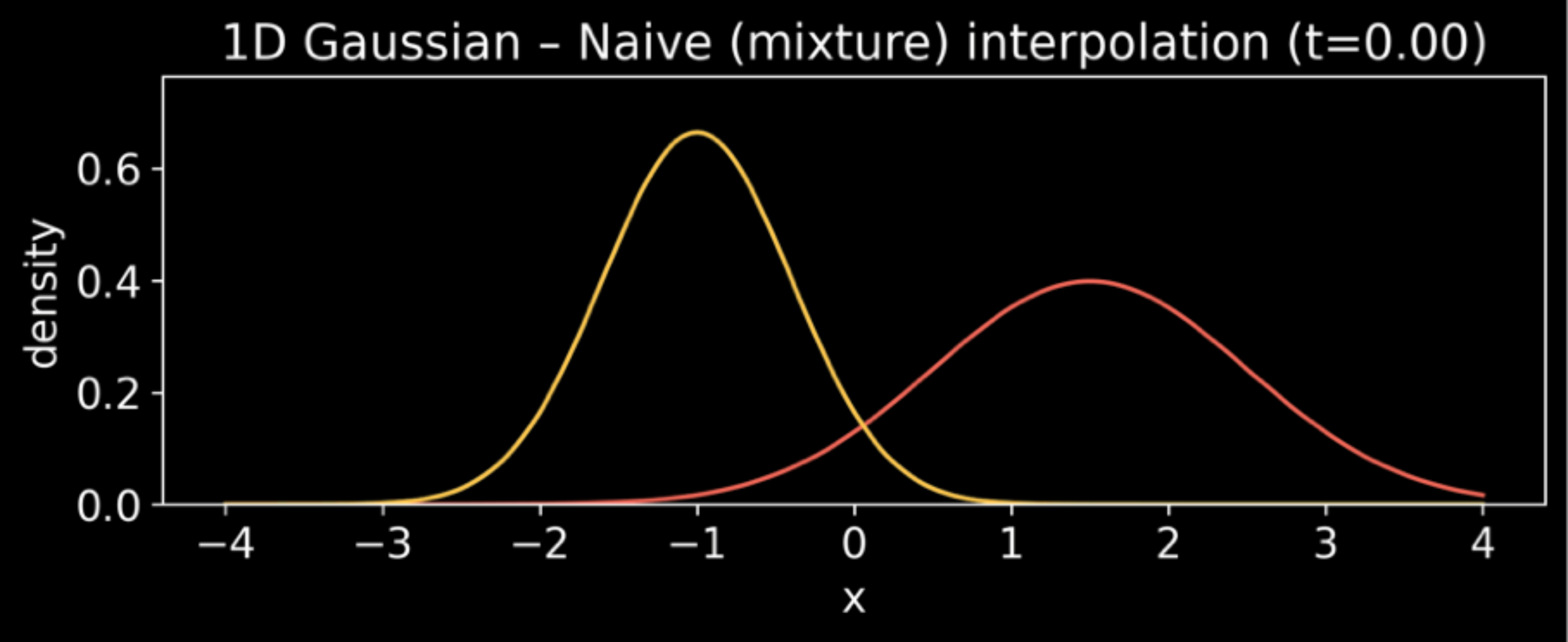
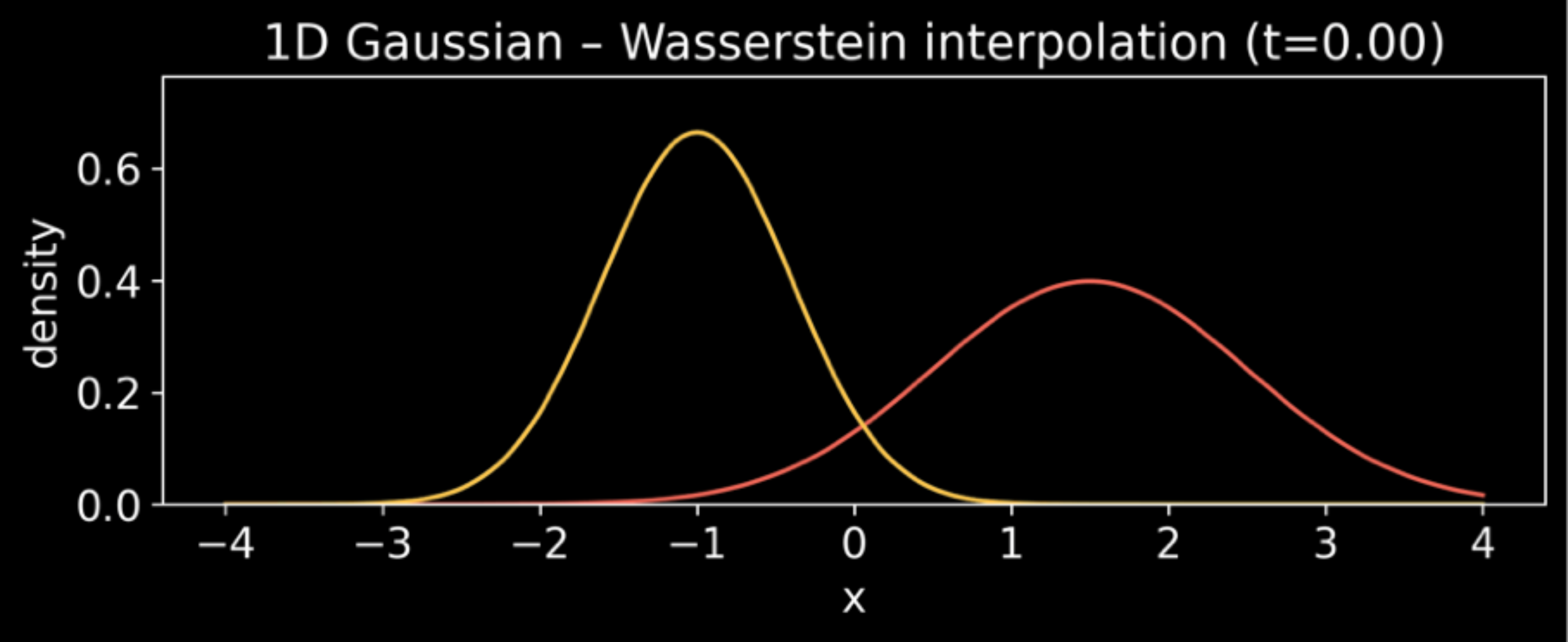
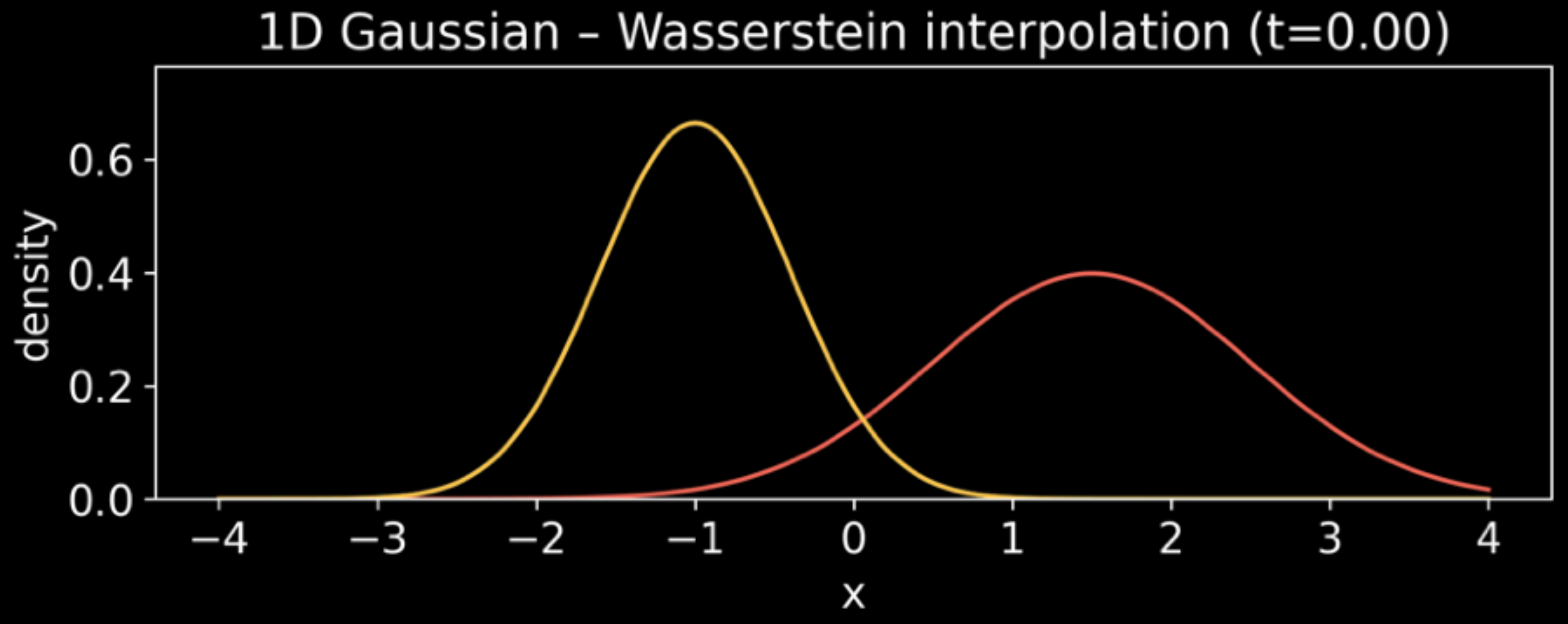
$$\alpha = \mathcal{N}(m_0, \sigma_0) \quad \beta = \mathcal{N}(m_1, \sigma_1)$$

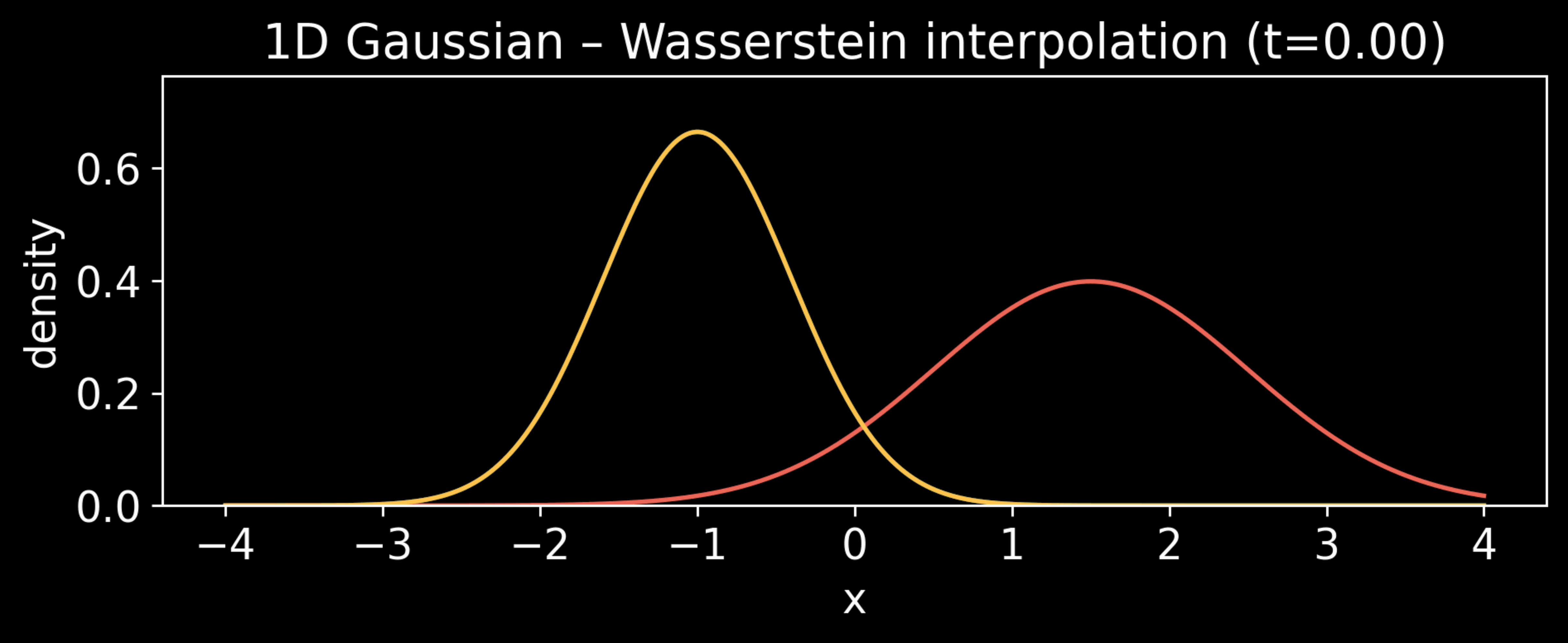
$$m_t = (1 - t)m_0 + tm_1$$

$$\sigma_t = (1 - t)\sigma_0 + t\sigma_1$$

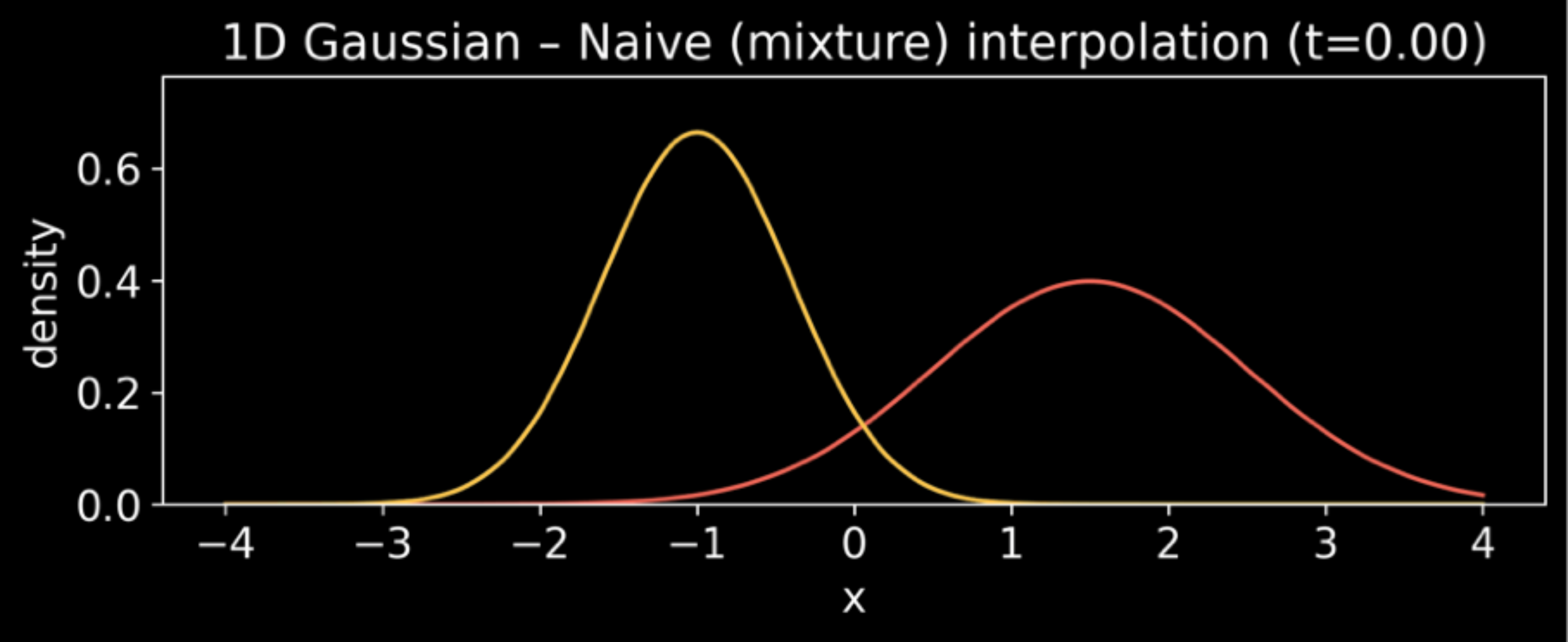

1D Gaussian – Naive (mixture) interpolation (t=0.00)
density
0.0
0.2
0.4
0.6
−4
−3
−2
−1
0
1
2
3
4
x

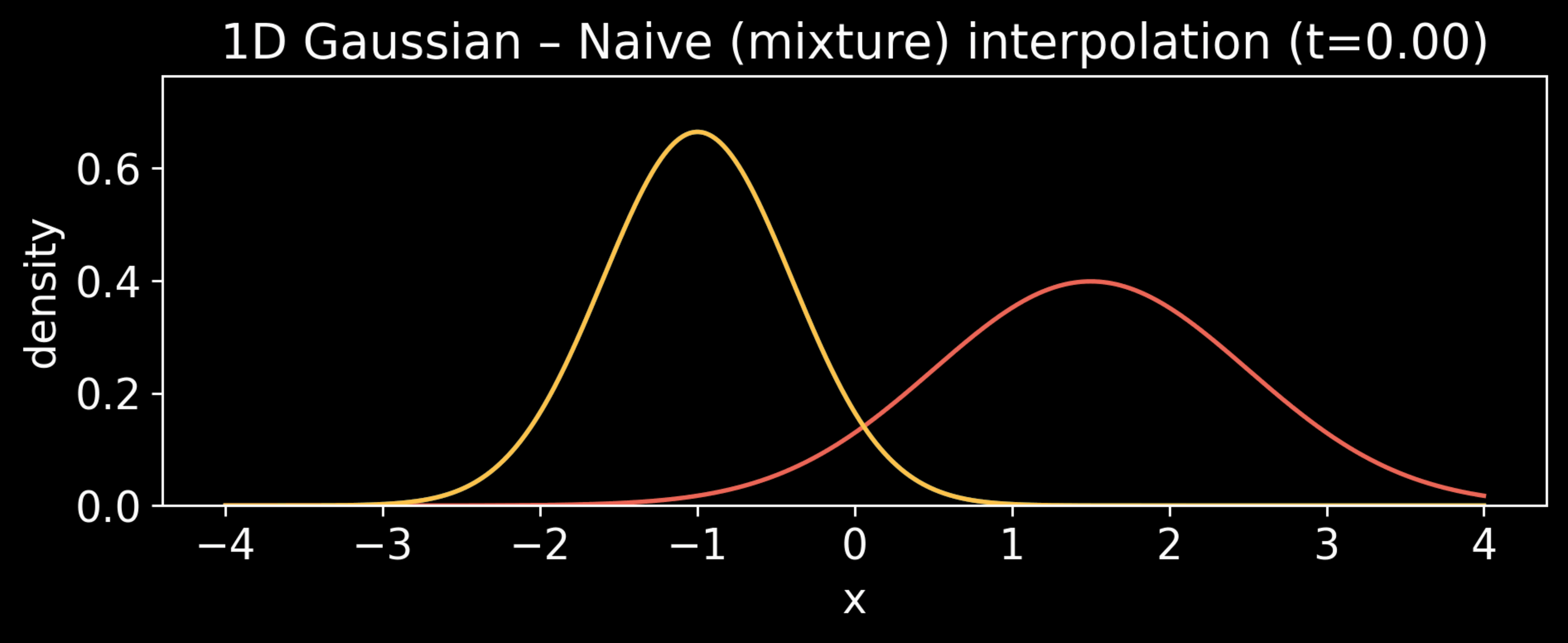

1D Gaussian – Wasserstein interpolation (t=0.00)
density
0.0
0.2
0.4
0.6
−4
−3
−2
−1
0
1
2
3
4
x


1D Gaussian – Wasserstein interpolation (t=0.00)
density
0.0
0.2
0.4
0.6
−4
−3
−2
−1
0
1
2
3
4
x



1D Gaussian – Wasserstein interpolation (t=0.00)
density
0.0
0.2
0.4
0.6
−4
−3
−2
−1
0
1
2
3
4
x


1D Gaussian – Naive (mixture) interpolation (t=0.00)
density
0.0
0.2
0.4
0.6
−4
−3
−2
−1
0
1
2
3
4
x


1D Gaussian – Naive (mixture) interpolation (t=0.00)
density
0.0
0.2
0.4
0.6
−4
−3
−2
−1
0
1
2
3
4
x

Example: Gaussian → Gaussian

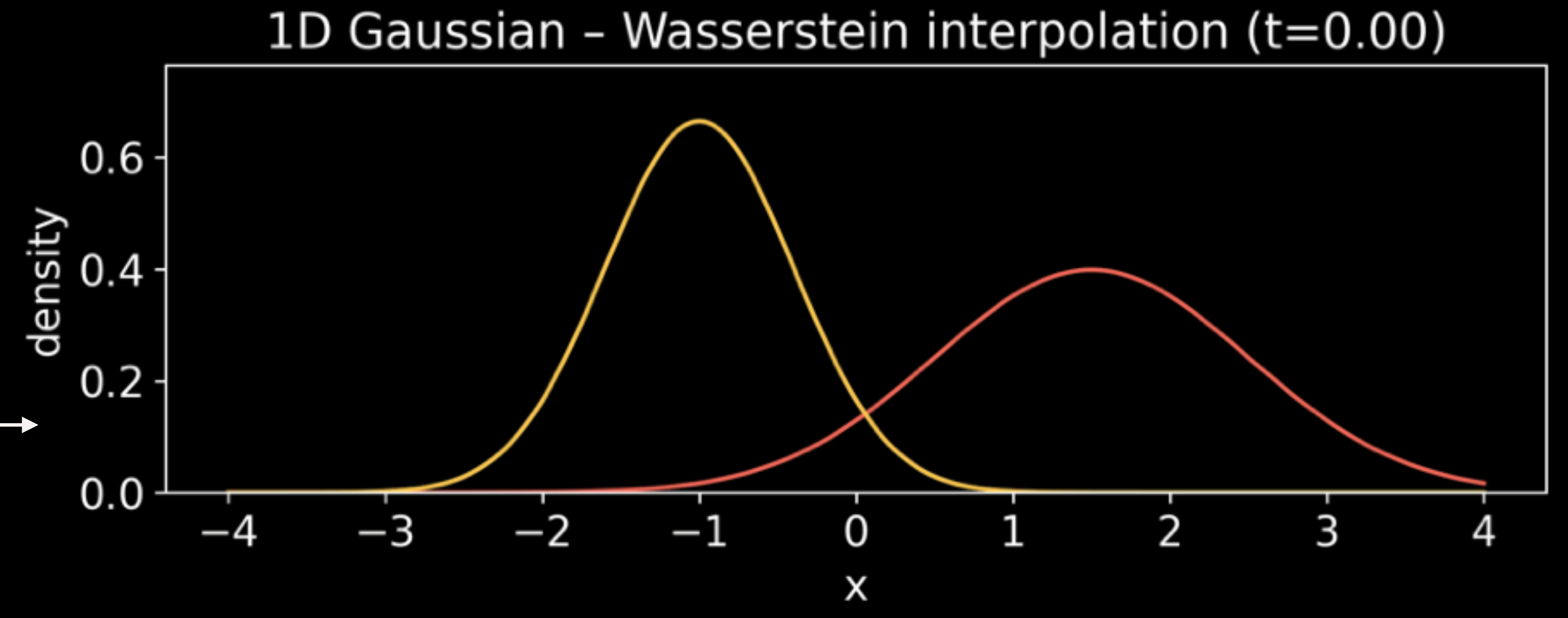
$\alpha = \mathcal{N}(m_0, \sigma_0)$, $\beta = \mathcal{N}(m_1, \sigma_1)$

Mean evolves linearly:

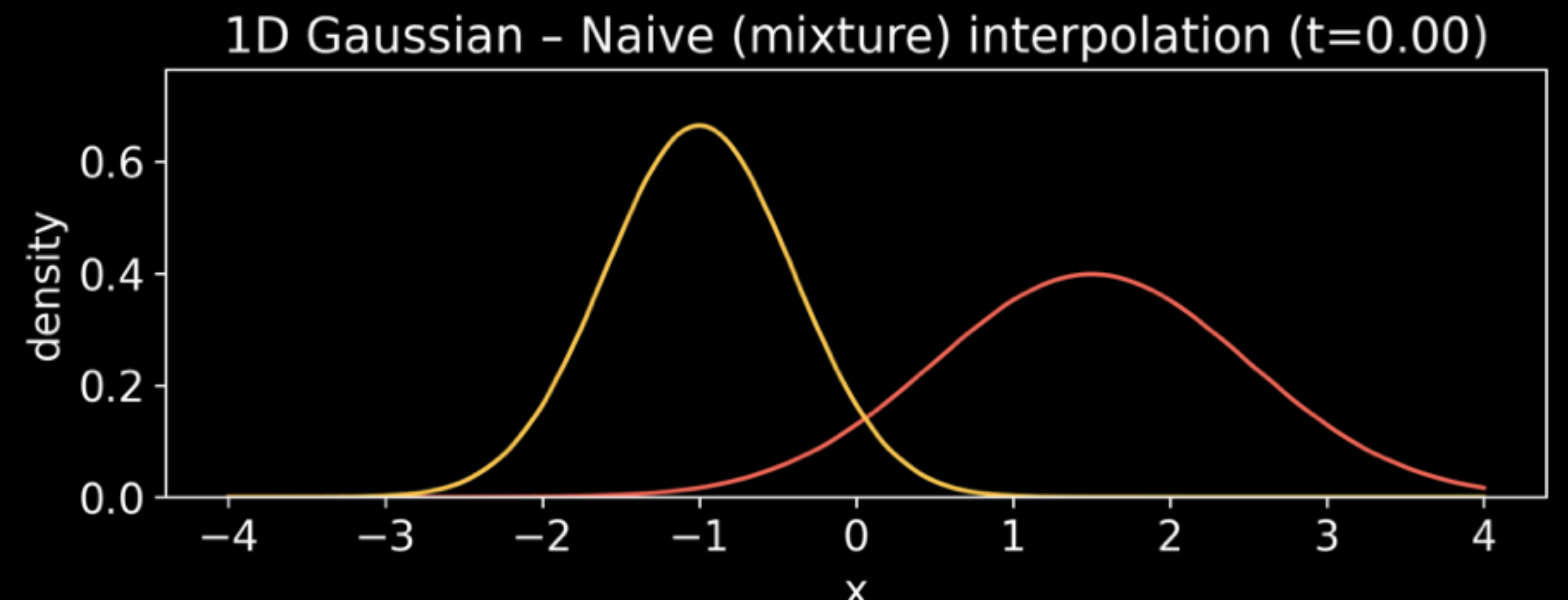
$$m_t = (1 - t)m_0 + tm_1$$

Standard deviation evolves linearly:

$$\sigma_t = (1 - t)\sigma_0 + t\sigma_1$$



very different dynamics from mixture interpolation:



Example: Gaussian → Gaussian

$$\alpha = \mathcal{N}(m_0, \Sigma_0), \beta = \mathcal{N}(m_1, \Sigma_1)$$

Mean evolves linearly:

$$m_t = (1-t)m_0 + tm_1$$

Covariance evolves nonlinearly:

$$\Sigma_t = \big((1-t)I + tA\big)\Sigma_0\big((1-t)I + tA\big)^T$$